DRAFT STANDARD OPERATING PROCEDURE

Sugarcane Crossing Block Layout and Management

Sugarcane Breeding and Germplasm Conservation Program

SOP No.	SCB-BR-SOP-03	**Revision**	0
Effective date	_____	**Supersedes**	New
Prepared by	_____	**Reviewed by**	_____
Approved by	_____	**Controlled copy**	Yes / No

Status: Draft technical SOP. Field dimensions, thresholds, chemical treatments, quarantine steps, and local compliance requirements must be validated and approved by the implementing institution before adoption.

1. Purpose

To establish and manage a crossing/flowering block that provides healthy, correctly identified, genetically diverse parents with predictable access for flowering surveillance, pollen testing, arrow collection, and controlled mating.

2. Scope

Covers field crossing blocks, photoperiod-derived parents transferred for crossing, parent holding blocks, and backup parental material. It does not replace the separate controlled pollination SOP.

3. Parent population strategy

- Core elite parents with demonstrated breeding value or combining ability.
- New or experimental parents requiring progeny testing.
- Trait donors for disease, insect, abiotic stress, maturity, quality, ratooning, or architecture.
- Basic germplasm and introgression materials used to broaden the genetic base.
- Backup/holding parents whose offspring are under evaluation or whose flowering is uncertain.

4. Pre-plant requirements

- Confirmed accession/parent identity and pedigree.
- Plant-health clearance and clean seed cane.
- Crossing objective and intended parent-use classification.
- Field selected for irrigation/drainage, access, security, and ability to isolate or manage pollen movement.
- Digital field map prepared before planting; permanent labels available.

5. Layout principles

Group parents by expected flowering behavior or management requirement rather than randomly scattering them across the station. A practical structure is early-, mid-, and late-flowering sections, with separate introgression/wild and parent-holding sections. Exact row length, row spacing, and stool number must follow local agronomy and the station's flower-production needs.

6. Suggested functional block arrangement

Block	Primary contents	Management purpose
A	Early flowering parents	Early surveillance and synchronized crossing
B	Mid-season flowering parents	Main crossing population
C	Late flowering parents	Late crossing opportunities / backups
D	Wild / introgression / special donors	Containment and specialized handling
E	Parent holding / backup	Redundancy and future use

7. Field layout and identification

1. Prepare the block map before planting. Assign immutable field coordinates using Block-Row-Position or Block-Row-Plot coding.
2. Plant enough stools of each important parent to provide multiple arrows and backup material. Very high-value parents should have duplicate plants outside the main crossing block.
3. Install durable labels at the beginning and end of each parent row/plot. Labels must show parent/accession ID and field coordinate; use barcode/QR where practical.
4. Provide access alleys wide enough for repeated inspection, safe arrow cutting, and movement of carts or personnel without damaging neighboring parents.
5. Use border or buffer rows where locally needed for field protection, wind reduction, or operational separation.

6. Do not plant unidentified surplus material in empty positions within the block. Any replacement planting must be documented and relabeled.
7. Update the digital map immediately after any mortality, replacement, relocation, or identity correction.

8. Flowering surveillance

During the expected induction and emergence period, inspect parents at least 2-3 times per week; increase to daily or near-daily observation when arrows are emerging and crossing activity is active. Sugar Research Australia reports inspecting up to hundreds of candidate parents several times per week and barcoding individual flowers in its operational breeding system.

9. Minimum flowering observations

- Date of first flag-leaf / boot indication if used by the program.
- Date and percentage of arrow emergence.
- Number of usable arrows per parent.
- Anther opening/anthesis stage.
- Pollen viability/fertility result for the current crossing session.
- Arrow health, damage, sterility, or contamination concerns.
- Disposition: female, male, backup, not used, or rejected.

10. Parent-use decisions

Do not permanently classify a clone as male or female based only on historical labels. Pollen fertility can vary with environment. As an operational example, Sugar Research Australia tests pollen each crossing session and uses >20% viable pollen as a male threshold in its system; institutions should validate their own threshold and staining method before adoption.

11. Quality-control checkpoints

- Map-to-label agreement verified before flowering season.
- No two parents share the same field coordinate or barcode.
- Backup stools exist for high-value parents.
- Flowering records are date-stamped and linked to the specific parent/arrow.
- Pedigree conflicts are resolved before a parent is used in a high-priority cross.
- Diseased or suspect parents are withheld according to plant-health rules.

12. Crossing block inspection sheet

Parent ID	Field position	Flower stage	Usable arrows	Pollen test	Provisional sex	Action / remarks

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References and technical basis

Sugar Research Australia - Producing New Sugarcane Varieties - Web source

Sugar Research Australia - Variety development update, August 2026 - Web source

ICAR-Sugarcane Breeding Institute - germplasm and breeding resources - Web source